Question - 2

(i) $\mathcal{L}$ to $\hat{y}$:

$$\mathcal{L} = -[y \log \hat{y} + (1-y) \log (1-\hat{y})]$$

$$\frac{\partial \mathcal{L}}{\partial \hat{y}} = -\left[\frac{y}{\hat{y}} - \frac{1-y}{1-\hat{y}}\right] = \frac{\hat{y}-y}{\hat{y}(1-\hat{y})}$$

(ii) $\hat{y}$ to s $\quad \hat{y} = \sigma(s)$

$$\frac{\partial \hat{y}}{\partial s} = \sigma(s)(1-\sigma(s))$$

$$= \hat{y}(1-\hat{y})$$

(iii) from s to (+b):

$$s = (w^T x) + b$$

$$\frac{\partial s}{\partial (w^T x)} = 1 \quad \& \quad \frac{\partial s}{\partial b} = 1$$

(iv) from dot to $(w^T x)$

$$\frac{\partial s}{\partial w} = x^T, \quad \frac{\partial s}{\partial x} = w^T$$

Deriving:

$$\frac{\partial \mathcal{L}}{\partial s} = \frac{\partial \mathcal{L}}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial s}$$

$$\frac{\partial \mathcal{L}}{\partial s} = \frac{\hat{y}-y}{\hat{y}(1-\hat{y})} \cdot \hat{y}(1-\hat{y})$$

$$\frac{\partial \mathcal{L}}{\partial s} = \hat{y} - y$$

② $x = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ $w = \begin{bmatrix} 0.5 \\ -1 \end{bmatrix}$, $b = 0$, $y = 1$

forward pass:

s: $w^T x = (0.5)(1) + (-1)(2) = 0.5 - 2 = -1.5$

$s = w^T x + b = -1.5 + 0 = -1.5$

$\hat{y}$: $\hat{y} = \sigma(-1.5) = \frac{1}{1 + e^{-(-1.5)}} = \frac{1}{1 + e^{1.5}}$

$= 0.1824$

L: $L = -[1 \cdot \log(0.1824) + 0]$

$= -\log(0.1824) \approx 1.7015$